9 Natalia pays £13 995 for a car.
Lauren pays £14 495 for a car.

Assume that

the rate of depreciation for Natalia's car is 12% per annum
and the rate of depreciation for Lauren's car is 13% per annum.

(a) Work out whose car will have the greater value at the end of 3 years.
You must show all your working.

Natalia keeps 100% − 12% = 88% each year.

1 mark
One-year multiplier

Natalia: £13995 × 0.88^3 = £9537.20064.

1 mark
Three annual multipliers

Lauren: £14495 × 0.87^3 = £9545.000985.

1 mark
Calculate both cars

Lauren: £9545.00 > £9537.20, so her car is worth more.

1 mark
Compare both values

(4)

The rate of depreciation assumed for Natalia's car was too low.

(b) How does this affect the value of her car at the end of 3 years?

A higher depreciation rate leaves a lower value.

1 mark
Worth less

Natalia's car will be worth less than £9537.20.

(1)

(Total for Question 9 is 5 marks)